

GR 2 Lecture Note

Yu Liu
March 05, 2026

Abstract

This is the lecture note of the GR 2 course in EPFL 2026 spring semester. Mainly covering the topics in cosmology

Contents

1	Lecture 1: FRW Metric Spactime	2
1.1	Space Slices Metric	2
1.1.1	S^3 3-sphere	2
1.1.2	$\mathbb{H}^3$ 3-hyperbolic space	3
1.1.3	Unified Form of Spacial Metric	3
1.2	FRW Metric	3
1.2.1	Definition in Comoving Coordinates	3
1.2.2	Unit of the Scale Factor	3
1.2.3	Remark on Comoving Coordinates	4
1.2.4	Connection and Curvature	4
2	Lecture 2: Friedmann Equations and Solutions	5
2.1	Einstein Field Equations	5
2.1.1	Matter Energy-Momentum Tensor in Comoving Frame	5
2.1.2	Friedmann Equations	5
2.1.3	Energy Conservation of Matter	6
2.2	Matter Content and Solutions	6
2.2.1	Matter Content and Equation of State	6
2.2.2	Solution 1: Einstein Static Universe	7
2.2.3	Solution 2: Flat Matter Dominated Universe	7
2.2.4	Solution 3: Flat General Matter Dominated Universe	7
3	Lecture 3: Cosmological Horizons, dS Spacetime and Lengths	9
3.1	Cosmological Horizons	9
3.1.1	Case 1 : $\omega > -\frac{1}{3}$ decelerating universe	9
3.1.2	Case 2: $\omega < -\frac{1}{3}$ accelerating universe	9
3.2	De Sitter Spacetime	10
3.2.1	Cosmological Constant Case with Flat Spatial Curvature	10
3.2.2	Penrose Diagram	10
3.2.3	Global de Sitter Spacetime	11
3.3	Hubble Parameter	12
3.4	Lengths in Cosmology	13
3.4.1	Redshift	13
3.4.2	Physical Distance and Coordinate Distance	14
3.4.3	Luminosity Distance	14
	Bibliography	15

1 Lecture 1: FRW Metric Spactime

For a cosmological model, we need to make some assumptions about the universe. The most common assumption is the cosmological principle:

Theorem 1.1

Cosmological Principle

The universe is homogeneous and isotropic on large scales.

The cosmological principle strongly constains the form of the metric. We assume that we can select a space slice at each time in some coordinate system, and consider the induced metric on each time slice.

1.1 Space Slices Metric

The cosmological principle requires that the metric on each time slice is homogeneous and isotropic, which means that the metric on each time slice is a maximally symmetric space. A maximally symmetric space is a space that has the maximum number of symmetries, which means that the Riemann curvature tensor has the following form:

$$\begin{aligned} R^{(3)} &= \text{constant} \\ R_{ij} &= \frac{1}{3} R^{(3)} g_{ij} \\ R_{ijkl} &= \frac{1}{6} R^{(3)} (g_{ik} g_{jl} - g_{il} g_{jk}) \end{aligned} \tag{1}$$

Thus we can see that a spactial metric can be classified into three types, and in fact each type of metric can be written in a simple form:

- $R^{(3)} = 0$, the metric is $\mathbb{R}^3$ 3-plane
- $R^{(3)} > 0$, the metric is $\mathbb{S}^3$ 3-sphere
- $R^{(3)} < 0$, the metric is $\mathbb{H}^3$ 3-hyperbolic space (Euclidean AdS space)

The metirc of $\mathbb{R}^3$ is trivial, we will consider the other two cases in the following:

1.1.1 $\mathbb{S}^3$ 3-sphere

The metric of a 3-sphere can be given by inducing from a 4-dimensional Euclidean space with Cartesian coordinates x^i and the metric $ds^2 = \delta_{ij} dx^i dx^j$. We can induce a coordinate and metric by the following constaint and parametrization:

- Constraint: $\sum_i x_i^2 = a^2$
- Parametrization:

$$\begin{aligned} x_1 &= R \sin \theta \cos \varphi \\ x_2 &= R \sin \theta \sin \varphi \\ x_3 &= R \cos \theta \\ x_4 &= \sqrt{a^2 - R^2} \end{aligned} \tag{2}$$

The induced metric is given by:

$$ds^2 = \frac{a^2 dr^2}{a^2 - r^2} + r^2 d\Omega_2^2 \tag{3}$$

We can calculate the Ricci Scalar of this metric, and we find that $R^{(3)} \sim \frac{1}{a^2}$.

1.1.2 $\mathbb{H}^3$ 3-hyperbolic space

Similarly we can embed the 3-hyperbolic space into a 4-dimensional Minkowski space with Cartesian coordinates x^i to get the metric. Another easy way of seeing the result is simply take the metric of the 3-sphere and replace a^2 with $-a^2$. This will definitely give us a solution to the constraints Equation 1 with negative curvature. Thus we can write the metric of the 3-hyperbolic space as:

$$ds^2 = \frac{a^2 dr^2}{a^2 + r^2} + r^2 d\Omega_2^2 \quad (4)$$

1.1.3 Unified Form of Spacial Metric

With above preparations, we can write three types of metrics in a unified form:

$$ds^2 = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_2^2 \right) \quad (5)$$

Here $k = -1, 0, 1$ stands for the three types of metrics. $k = 1$ gives $\mathbb{S}^3$ and $k = -1$ gives $\mathbb{H}^3$.

1.2 FRW Metric

1.2.1 Definition in Comoving Coordinates

With the spacial slice metric prepared, we can add a time coordinate to make it a full 4 dimensional metric ansatz. A typical form of the metric is given by:

Definition 1.2

FRW Metric

The FRW metric is given by:

$$\begin{aligned} ds^2 &= -dt^2 + a^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_2^2 \right) \\ &= -dt^2 + a^2(t) \gamma_{ij} dx^i dx^j \end{aligned} \quad (6)$$

Mathematically, we can rigorously show that the FRW metric is the only metric that satisfies the cosmological principle. Physically, we can also understand that the FRW metric is the most general metric that can describe a homogeneous and isotropic universe.

Remark

However, we shall notice that such metric is only defined on a patch of the manifold. Different manifold with different topology can have the same local metric. Thus, the global structure of the spacetime is undetermined.

1.2.2 Unit of the Scale Factor

We can see that:

- For $k \neq 0$;, the scale factor $a(t)$ has the physical meaning of radius of spacial curvature. Thus, we can assign it a unit of length and take the spacial coordinate dimensionless.
- For $k = 0$: the scale factor $a(t)$ can be taken as dimensionless, and the spacial coordinate can be taken as having a unit of length.

In fact in this case, at fixed time, $a(t = t_0)$ is meaningless, what physical is the ratio $\frac{a(t)}{a(t_0)}$.

1.2.3 Remark on Comoving Coordinates

[Definition 1.2](#) is certainly written in a certain coordinate system, which is called the **comoving coordinates**. This means that the particles at rest in this frame (at rest means at rest spacially and moving in the time coordinate) are free (which means they move in geodesics) We can see that indeed $u^0 = 1$ is a solution to the geodesic equation.

1.2.4 Connection and Curvature

We can calculate the Levi-Civita connection and the curvature of the FRW metric in the comoving coordinates:

- Metric:

$$g_{00} = -1, \quad g_{ij} = a^2 \gamma_{ij} \quad g^{00} = -1, \quad g^{ij} = a^{-2} \gamma^{ij} \quad (7)$$

- Curvature:

$$\begin{aligned} R_{00} &= -3 \frac{\ddot{a}}{a} \quad R_{0i} = 0 \\ R_{ij} &= (a\ddot{a} + 2\dot{a}^2 + 2k) \gamma_{ij} \\ R &= 6 \frac{a\ddot{a} + \dot{a}^2 + k}{a^2} \end{aligned} \quad (8)$$

where γ_{ij} is the spacial metric. We can see that the curvature is determined by the scale factor $a(t)$ and the curvature parameter k .

2 Lecture 2: Friedmann Equations and Solutions

We can rewrite the Einstein Field Equations with two ansatz:

- The metric is given by the Friedmann-Robertson-Walker (FRW) metric
- The EM tensor of matter is given by a perfect fluid in the comoving frame

2.1 Einstein Field Equations

We consider the general Einstein Field Equation with a cosmological constant:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi GT_{\mu\nu} \quad (9)$$

2.1.1 Matter Energy-Momentum Tensor in Comoving Frame

In Section 1.2.4 we have calculated the curvature tensor in the comoving frame, which is the LHS of Equation 9. Now we need to calculate the EM tensor of matter in the comoving frame. We focus on:

- The matter is a perfect fluid and being at rest in the comoving frame

which means that the EM tensor has the following form in the comoving frame:

$$T_{\mu\nu} = u_\mu u_\nu (\rho + p) + pg_{\mu\nu} \quad (10)$$

And we assume that the matter is at rest in the comoving frame, which means that the 4-velocity is given by $u^\mu = (1, 0, 0, 0)$. Thus we have the only non-zero component of the EM tensor is:

$$T_{00} = \rho, \quad T_{ij} = pg_{ij} \quad \text{or} \quad T^{00} = \rho, \quad T^{ij} = pg^{ij} = p \frac{\gamma^{ij}}{a^2} \quad (11)$$

Generally, the homogeneity and isotropy of the universe requires that the EM tensor $\rho(t), p(t)$ only depends on time. This suits that the metric is also only depends on time.

Normally, we assume that there is a relation between the energy density and pressure, depend on the matter itself. Thus, the EM tensor has only one degree of freedom, which is the energy density ρ .

2.1.2 Friedmann Equations

Now we can plug in the curvature tensor and the EM tensor into the Einstein Field Equation, and we can get the 00 component of the Einstein Field Equation, which is called the Friedmann Equation:

Theorem 2.1

Friedmann Equation

The Friedmann Equation is given by the 00 component of the EFE of a rest perfect fluid in the comoving frame, which is given by:

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} \quad (12)$$

Let's now count the variable and equations, usually we fix a k for the model we want to consider. Then, the EM Tensor have 1 DoF $\rho(t)$, the metric have 1 DoF $a(t)$, and we have 1 equation, which is the Friedmann Equation. Thus we need another equation to solve the system.

We first consider the other Terms of the Einstein Field Equation, which is the ij component. We can get another equation from the ij component of the EFE, which is given by:

Theorem 2.2

Spacial Component of EFE

The ij component of the EFE of a rest perfect fluid in the comoving frame is given by:

$$2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = -8\pi Gp + \Lambda \quad (13)$$

Remark

We only have one spacial equation, this is due to the fact that the spacial metric is dependent on γ_{ij} up to one factor. Thus, the spacial component of the EFE only gives one equation.

Sometimes we combine the two equations together, and we can get rid of the term $\dot{a}$, which is :

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3} \quad (14)$$

From this equation we can easily see when the universe is accelerating or decelerating. If we assume that $\Lambda = 0$ then:

- If $\rho + 3p > 0$ then the universe is decelerating $\ddot{a} < 0$
- If $\rho + 3p < 0$ then the universe is accelerating $\ddot{a} > 0$

2.1.3 Energy Conservation of Matter

Another commonly used choice of the second equation is the energy conservation of matter. Note that this equation is not independent of EFE, since it is given by EFE and the Bianchi Identity.

Theorem 2.3

Energy Conservation of Matter

The energy conservation of matter is given by:

$$\partial_t(\rho a^3) + p\partial_t a^3 = 0 \quad (15)$$

We can see that the energy conservation is equivalent to the first law of thermodynamics, if we view the matter as a thermodynamic system with volume $V = a^3$ and internal energy $U = \rho a^3$.

2.2 Matter Content and Solutions

Now we try to find solutions to the EFE. But we need to specify the matter content exists in the universe.

2.2.1 Matter Content and Equation of State

We usually focus on Model behave as

$$P = \omega\rho \quad (16)$$

And usually the coefficient ω is given by:

- Pressureless Dust: $\omega = 0$
- Relativistic Fluid (Radiation): $\omega = \frac{1}{3}$
- Cosmological Constant: $\omega = -1$

We believe that $\omega \geq -1$ and cc is the lower bound, which is called the null energy condition.

Remark

We can see from the EFE that a matter with $\omega < -1$ gives out $\rho = \text{const}$ in the energy conservation of matter, this is equivalent to having a cosmological constant of $\Lambda = 8\pi G\rho$.

Btw, we may assume that $\rho > 0$ the matter have positive energy density. Thus, the “cosmological constant” case is in fact “positive cosmological constant” case.

2.2.2 Solution 1: Einstein Static Universe

If we assume that $\rho \neq 0$, $p = 0$; $\dot{a} = 0$. which means that the universe is full of pressureless dust and is static. Then the two EFE becomes:

$$\frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \frac{k}{a^2} = \Lambda \quad (17)$$

If we take a positive cosmological constant, then we can get the following solution:

$$k = 1 \quad a = \sqrt{\frac{1}{\Lambda}} \quad \rho = \frac{\Lambda}{4\pi G} \quad (18)$$

This solution is called the Einstein Static Universe, which is a static universe with positive curvature and a positive cosmological constant. However, this is unrealistic

2.2.3 Solution 2: Flat Matter Dominated Universe

We assume that $k = 0$, $\Lambda = 0$, and the universe is full of pressureless dust. Then the Friedmann Equation and the Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \partial_t(\rho a^3) = 0 \quad \text{or} \quad \rho a^3 = C \quad (19)$$

we can get a differential equation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \frac{C}{a^3} \quad (20)$$

The solution to this equation is given by:

$$a(t) = a_0(t - t_0)^{\frac{2}{3}} \quad \rho \sim \frac{1}{t^2} \quad (21)$$

We can notice that this solution $\dot{a} > 0$ and $\ddot{a} < 0$, which means that the universe is expanding and decelerating.

2.2.4 Solution 3: Flat General Matter Dominated Universe

Consider a more general case with $k = 0$, $\Lambda = 0$, and the matter content is given by a general equation of state $P = \omega\rho$. This time the Friedmann Equation and the Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + p) = 0 \quad (22)$$

This gives us a solution:

$$\rho = Ca^{-3(1+\omega)}, \quad a(t) = a_0(t - t_0)^{\frac{2}{3(1+\omega)}} \quad (23)$$

There are some special properties of this solution:

- **Null Energy Condition** : Notice that we believe that $\omega \geq -1$. Thus we can see that $a(t)$ will always have a singularity at $t = t_0$, for the metric in this coordinate.
- **Matter density time dependent** : We can see that if one kind of matter dominates the universe, then:

$$\rho \sim \frac{1}{t^2} \quad (24)$$

And if we have many kinds of matter and the effective ω is given by ω , then matter ρ_i evolve as:

$$\rho_i \sim t^{-2\frac{1+\omega_i}{1+\omega}} \quad (25)$$

- **Accelerating and Decelerating** : we can calculate $\ddot{a}$ and we see that:
 - If $\omega > -\frac{1}{3}$ then the universe is decelerating $\ddot{a} < 0$
 - If $\omega < -\frac{1}{3}$ then the universe is accelerating $\ddot{a} > 0$
- **Spacial Curvature as Matter**: An interesting observation is that having non-zero spacial curvature k is equivalent as choosing a matter of:

$$\omega = -\frac{1}{3} \quad (26)$$

Thus in fact we can view the spacial curvature as a kind of matter content in the universe. We can also see that this matter makes the universe not accelerating nor decelerating, $\ddot{a} = \text{const.}$

3 Lecture 3: Cosmological Horizons, dS Spacetime and Lengths

3.1 Cosmological Horizons

We consider the Penrose diagram of a FRW metric, generally the metric can be written as:

$$ds^2 = -dt^2 + a(t)^2 dx^2 \quad (27)$$

Here we use dx^2 to represent the spacial metric. Then we can make a change of variables:

$$d\eta = \frac{dt}{a(t)} \quad (28)$$

Thus we can have the metric using (η, x) coordinate as:

$$ds^2 = a(\eta)^2 (-d\eta^2 + dx^2) \quad (29)$$

which is a conformally Minkowski metric. Then we need to fix the shape of the Penrose diagram. To do this we have to know what value can we take for η as t goes from $t_0 = 0$ to $t = +\infty$. We can integrate the relation Equation 28 and assume at $t = 0, \eta = \eta_*$:

$$\eta = \int_0^t \frac{dt'}{a(t')} + \eta_* \quad (30)$$

We assume that the space is flat and the spacetime is filled with $P = \omega\rho$ content. Then a general solution is:

$$a(t) \sim t^{\frac{2}{3(1+\omega)}} \quad (31)$$

Then we can see that $\omega > -\frac{1}{3}$ and $\omega < -\frac{1}{3}$ gives two cases for the behavior of η as t goes to zero and infinity.

3.1.1 Case 1 : $\omega > -\frac{1}{3}$ decelerating universe

In the case, we can find that the integral Equation 30 converge at $t = 0$ but diverge at $t = +\infty$. we can choose a suitable η_* to make η go from η_* to $+\infty$, and at $t = 0, \eta = \eta_*$ the metric is singular. Thus the penrose diagram behave as:

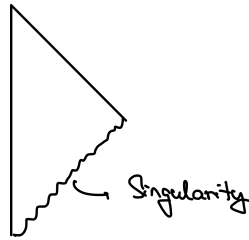


Figure 1: Penrose diagram for a decelerating universe with $\omega > -\frac{1}{3}$

3.1.2 Case 2: $\omega < -\frac{1}{3}$ accelerating universe

In this case, we find the integral Equation 30 diverge at $t = 0$ but converge at $t = +\infty$. This means that for an accelerating universe, we can choose a suitable η_* to make η go from $-\infty$ to a finite number and at $t = 0, \eta = -\infty$ there is a singularity. Thus the penrose diagram be like:

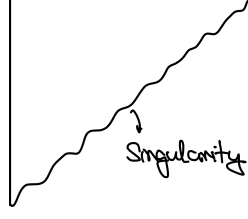


Figure 2: Penrose diagram for an accelerating universe with $\omega < -\frac{1}{3}$

We can see that in this case, there exist something called a **Cosmological Horizon**. It is originate from the fact that two points in the spacetime can never send information to each other, for the light can never reach each other before touching the future infinity.

This is interpret as the Universe is accerlating in expanding, thus the light can never catch up with the expansion of the universe.

3.2 De Sitter Spacetime

3.2.1 Cosmological Constant Case with Flat Spatial Curvature

We now focus on a special case of $k = 0, \Lambda = 0$ universe with the matter content of $\omega = -1$ which is the cosmological constant case. Or equivalently $\Lambda \neq 0, \rho = 0$. Note that it is not include in the general matter case Section 2.2.4, since the solution is distinct.

In this case, the Friedmann Equation and Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{\Lambda}{3}, \quad \rho = \text{const} := \frac{\Lambda}{8\pi G} \quad (32)$$

Thus we can get the solution that:

$$a(t) = \exp(Ht), \quad H = \sqrt{\frac{\Lambda}{3}}, \quad (33)$$

This solution we can see that t goes from $-\infty$ to $+\infty$. There is a singularity at $t = -\infty$ where $a(t) = 0$.

Remark

Here we take a convention that $a_0 = 1$ and in fact there are two solutions e^{Ht} and e^{-Ht} . We can choose the solution e^{Ht} since it is more physical, which means that the universe is expanding. The other solution e^{-Ht} is a contracting universe, and singularity at $t = +\infty$ where $a(t) = 0$. However, this solution is not physical since we believe that the universe is expanding, thus we can discard this solution.

3.2.2 Penrose Diagram

We can make a change of variable:

$$\eta = \int \frac{dt}{a(t)} = -\frac{e^{-Ht}}{H} \quad (34)$$

Thus we can get the metric in the (η, x) then the metric becomes:

$$ds^2 = \frac{a_0^2}{H^2 \eta^2} (-d\eta^2 + dx^2) \quad (35)$$

we can see the range of the parameter η :

- as $t \rightarrow \infty$ $\eta \rightarrow 0^-$ which is far future.
- as $t \rightarrow -\infty$ $\eta \rightarrow -\infty$ which is the past singularity.

Thus the penrose diagram of this spacetime is like:

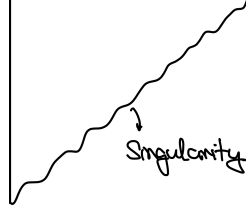


Figure 3: Penrose diagram for cosmological constant case with $\omega < -1$

which behaves just like the accerlating universe case in the general matter solution Figure 2.

3.2.3 Global de Sitter Spacetime

However, the singularity we get at $t = -\infty$ is not a real singularity!! These can be seen by the behavior of the curvature. We can prove that the curvature is a constant and don't have any singularity at $t = -\infty$. Thus, we expect we can do an extension of the spacetime to get rid of this coordinate singularity.

One approach to this is to investigate a seemingly different solution of the EFE, which is the **vaccum solution with $k = 1$** . The EFE reads:

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{1}{a^2} = \frac{\Lambda}{3}, \quad 2\left(\frac{\ddot{a}}{a}\right) + \left(\frac{\dot{a}}{a}\right)^2 + \frac{1}{a^2} = \Lambda \quad (36)$$

The solution to this equation is given by:

$$a(t) = H^{-1} \cosh(Ht), \quad H = \sqrt{\frac{\Lambda}{3}}, \quad (37)$$

This spacetime is called **Global dS Spacetime**.

$$ds^2 = -dt^2 + H^{-2} \cosh^2(Ht)(d\chi^2 + \sin^2(\chi)d\Omega_2^2) \quad (38)$$

We can see that the metric is non-singular at any t for the cosh function is well behaved and have no zero point. More importantly, we can prove:

- The cosmological constant case with flat spatial curvature is a patch of the global dS spacetime, which is called the **Poincare Patch**.

In penrose diagram, this can be shown as:

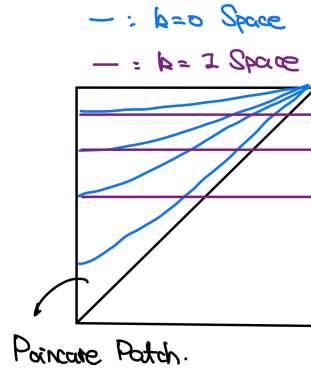


Figure 4: Penrose diagram for the global dS spacetime, where the blue region is the Poincare patch which is the cosmological constant case with flat spatial curvature

We can see that the blue line is a spacial slicing that make the spacial metric flat, yet the price to pay is that it only cover a patch of the global dS spacetime.

3.3 Hubble Parameter

We recap on the three equations we commonly use for solving EFE:

$$\begin{aligned} \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} &= \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \text{Friedmann Equation} \\ 2\left(\frac{\ddot{a}}{a}\right) + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} &= -8\pi Gp + \Lambda, \quad \text{Spacial EFE} \\ \dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + p) &= 0 \quad \text{Energy Conservation of Matter} \end{aligned} \tag{39}$$

If we consider a certain kind of matter $p_i = \omega_i \rho_i$, the enegy conservation gives us:

$$\rho_i = C a^{-3(1+\omega_i)}, \quad p_i = \omega_i \rho_i \tag{40}$$

Moreover, we can define a parameter:

Definition 3.1

Hubble Parameter

The **Hubble Parameter** is defined as:

$$H = \frac{\dot{a}}{a} \tag{41}$$

Thus the Friedmann Equation with $k = 0, \Lambda = 0$ can be written as:

$$H^2 = \frac{8\pi G}{3} \sum_i \rho_i \tag{42}$$

Further more we can define a parameter called **Critical Density** as:

Definition 3.2

Critical Density

The **Critical Density** is defined as:

$$\rho_c = \frac{3H_0^2}{8\pi G}, \quad \text{equivalently} \quad \rho_c = \sum_i \rho_i(t = t_0) \tag{43}$$

where H_0 is the Hubble Parameter at present time t_0 . And we can define Ω_i which is :

$$\Omega_i = \frac{\rho_i(t = t_0)}{\rho_c}, \quad \text{thus} \quad \sum_i \Omega_i = 1 \tag{44}$$

With all these preparation, we can rewrite the Friedmann Equation (with $k \neq 0, \Lambda \neq 0$ and with the energy conservation already used) as:

$$H^2 = H_0^2 \left(\Omega_\Lambda + \Omega_k \left(\frac{a_0}{a} \right)^2 + \Omega_m \left(\frac{a_0}{a} \right)^3 + \Omega_\gamma \left(\frac{a_0}{a} \right)^4 \right) \tag{45}$$

The first two term is the contribution from the cosmological constant and the spatial curvature, and the last two term is the contribution from the matter and the radiation, with certain ω_i determine its behavior.

If we rewrite LHS back with $\frac{\dot{a}}{a}$ and we group terms we can get:

$$\dot{a}^2 = H_0^2 \left(\Omega_\Lambda a^2 + \Omega_k a_0^2 + \Omega_m \frac{a_0^3}{a} + \Omega_\gamma \frac{a_0^4}{a^2} \right) \quad (46)$$

Which can be interpreted as the equation of motion of a particle with zero total energy moving in a potential:

$$U(a) = -H_0^2 \left(\Omega_\Lambda a^2 + \Omega_k a_0^2 + \Omega_m \frac{a_0^3}{a} + \Omega_\gamma \frac{a_0^4}{a^2} \right), \quad \dot{a}^2 + U(a) = 0 \quad (47)$$

3.4 Lengths in Cosmology

To investigate the cosmology, we need to define the lengths.

3.4.1 Redshift

Assume a process, a star emit a light at $t = t_1, r = r_1$ and the light is observed by us at $t = t_0, r = 0$. The light travel along null geodesics:

$$ds^2 = -dt^2 + a(t)^2 \frac{dr^2}{1 - kr^2} = 0 \quad \Rightarrow \quad dt = -a(t) \frac{dr}{\sqrt{1 - kr^2}} \quad (48)$$

There is a “-” sign since the light is traveling from r_1 to 0. Thus we can get evaluate a length of the light travel $l(r)$:

$$\int_{t_1}^{t_0} \frac{dt}{a(t)} = \int_0^{r_1} \frac{dr}{\sqrt{1 - kr^2}} = l(r_1) \quad (49)$$

We can calculate the length $l(r)$ for different spatial curvature:

$$l(r) = \begin{cases} \arcsin(r) & k = 1 \\ r & k = 0 \\ \operatorname{arcsinh}(r) & k = -1 \end{cases} \quad (50)$$

Then we can consider a process of two light signal emit at $t = t_1$ and $t = t_1 + \delta t_1$ and the light is observed at $t = t_0$ and $t = t_0 + \delta t_0$. Since the two light signal travel along the same null geodesics, we can get:

$$\frac{\delta t_0}{a(t_0)} = \frac{\delta t_1}{a(t_1)} \quad (51)$$

The frequency of the light is inversely proportional to the time interval $\omega \propto \frac{1}{t}$, thus we can get:

$$\frac{\omega_0}{\omega_1} = \frac{\delta t_1}{\delta t_0} = \frac{a(t_0)}{a(t_1)} \quad (52)$$

- Universe is Expanding: $a(t_0) > a(t_1)$ thus $\omega_0 > \omega_1$ which is called **Redshift**.
- Universe is Contracting: $a(t_0) < a(t_1)$ thus $\omega_0 < \omega_1$ which is called **Blueshift**.

The redshift can be described by the following quantity:

Definition 3.3

The **Redshift** is defined as:

$$z = \frac{\lambda_0 - \lambda_1}{\lambda_0} = \frac{a(t_0)}{a(t_1)} - 1 \quad (53)$$

note that t_0 is the time of getting the light and t_1 is the time of emitting the light.

Redshift

Usually, we know the λ_1 of many processes, and we measure λ_0 from experiments. Thus, we can use z as a measure of the distance.

3.4.2 Physical Distance and Coordinate Distance

For two comoving objects at $r = r_1$ and $r = r_2$, the coordinate distance is:

$$l_c = l(r_1) - l(r_2) \quad (54)$$

which is the distance separated in the spacial coordinate. However, its not the distance in the general metric. The physical distance is given by:

$$l_p = a(t)(l(r_1) - l(r_2)) = a(t)l_c \quad (55)$$

We know that for a comoving object, the spacial coordinate is fixed when moving in geodesics. Yet, the physical distance is time dependent due to the $a(t)$ parameter:

$$\frac{dl_p}{dt} = \dot{a}l_c = Hl_p, \quad H = \frac{\dot{a}}{a} \quad (56)$$

This is the **Hubble Law**, which fits well with the observation of the universe, which is one of the evidences for the expanding universe.

3.4.3 Luminosity Distance

We first define something as:

- **Standard Candle** : which is a object with known luminosity L , where $L = \frac{dE}{dt}$ is defined as the energy emitted by the object per unit time in the object's rest frame.

Then we consider a comoving telescope observing the standard candle, we can define a quantity called **Luminosity Distance** as:

Definition 3.4

Luminosity Distance

The **Luminosity Distance** is defined as:

$$P = L \frac{S}{4\pi d^2} \quad (57)$$

where P is the measured energy recieved per unit time by the telescope, and S is the area of the telescope. The luminosity distance d is defined as the distance that make the above equation hold.

Then we can calculate the luminosity distance and its relation with other quantities for a comoving telescope and star in the FRW metric set up.

Bibliography